

## Title: Ford GoBike System Data Exploration

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### INTRODUCTION

Ford GoBike is a bike-sharing transportation service that provides an affordable and convenient mode of travel for urban commuters. The system allows users to rent bicycles from one station and return them to another, promoting sustainable transportation and reducing traffic congestion.

This project focuses on performing Exploratory Data Analysis (EDA) on the Ford GoBike dataset for the year 2018. The analysis aims to understand customer behavior, trip duration patterns, user demographics, station popularity, and temporal trends. Through data visualization and statistical analysis, meaningful business insights and recommendations are generated to support data-driven decision-making.

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### OBJECTIVES

The primary objectives of this project are:

- To analyse bike-sharing usage patterns.
  - To identify customer behaviour and trip trends.
  - To understand demographic characteristics of riders.
  - To examine monthly, daily, and hourly usage patterns.
  - To identify popular stations and peak demand periods.
  - To generate business insights and recommendations.
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### TOOLS AND TECHNOLOGIES USED

| Tool                       | Purpose                 |
|----------------------------|-------------------------|
| Python                     | Data Analysis           |
| Pandas                     | Data Manipulation       |
| NumPy                      | Numerical Computations  |
| Jupyter Notebook / VS Code | Development Environment |
| CSV Files                  | Dataset Source          |

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## **DATASET DESCRIPTION**

The project utilizes twelve monthly CSV datasets containing Ford GoBike trip information for the year 2018.

### **Important Fields Used**

- Duration (duration\_sec)
  - Start Time
  - End Time
  - User Type
  - Member Gender
  - Member Birth Year
  - Bike ID
  - Start Station Name
  - End Station Name
  - Bike Share for All Trip
  - Month
  - Day
  - Hour
  - Age
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## **METHODOLOGY**

The project was completed using the following steps:

1. Imported all monthly CSV datasets.
  2. Combined datasets into a single DataFrame.
  3. Performed data cleaning and preprocessing.
  4. Checked missing values and duplicate records.
  5. Converted date columns into datetime format.
  6. Created Month, Day, Hour, and Age variables.
  7. Performed exploratory data analysis.
  8. Generated visualizations and insights.
  9. Developed business recommendations.
  10. Prepared final report and presentation.
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# DATA CLEANING & PREPROCESSING

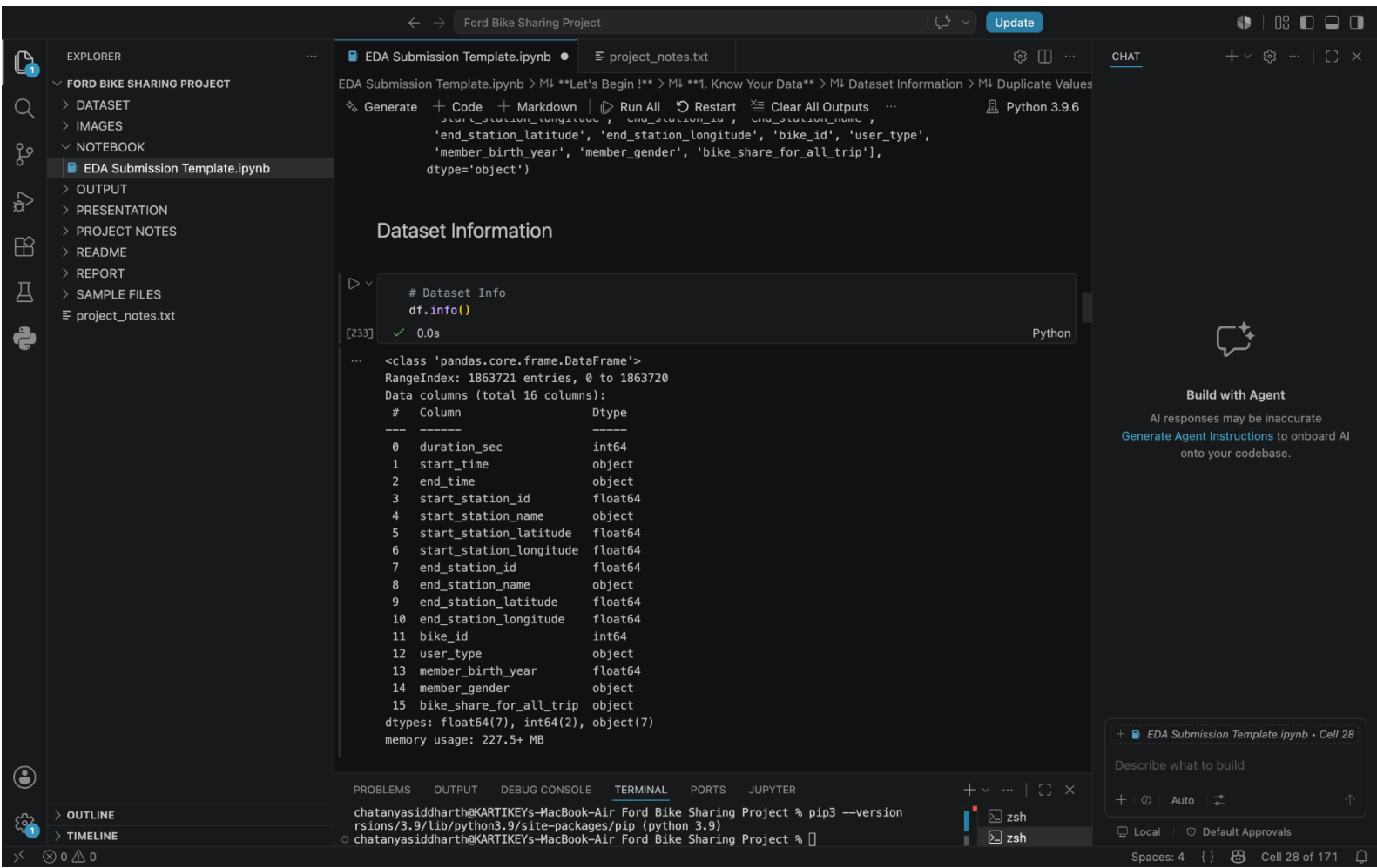
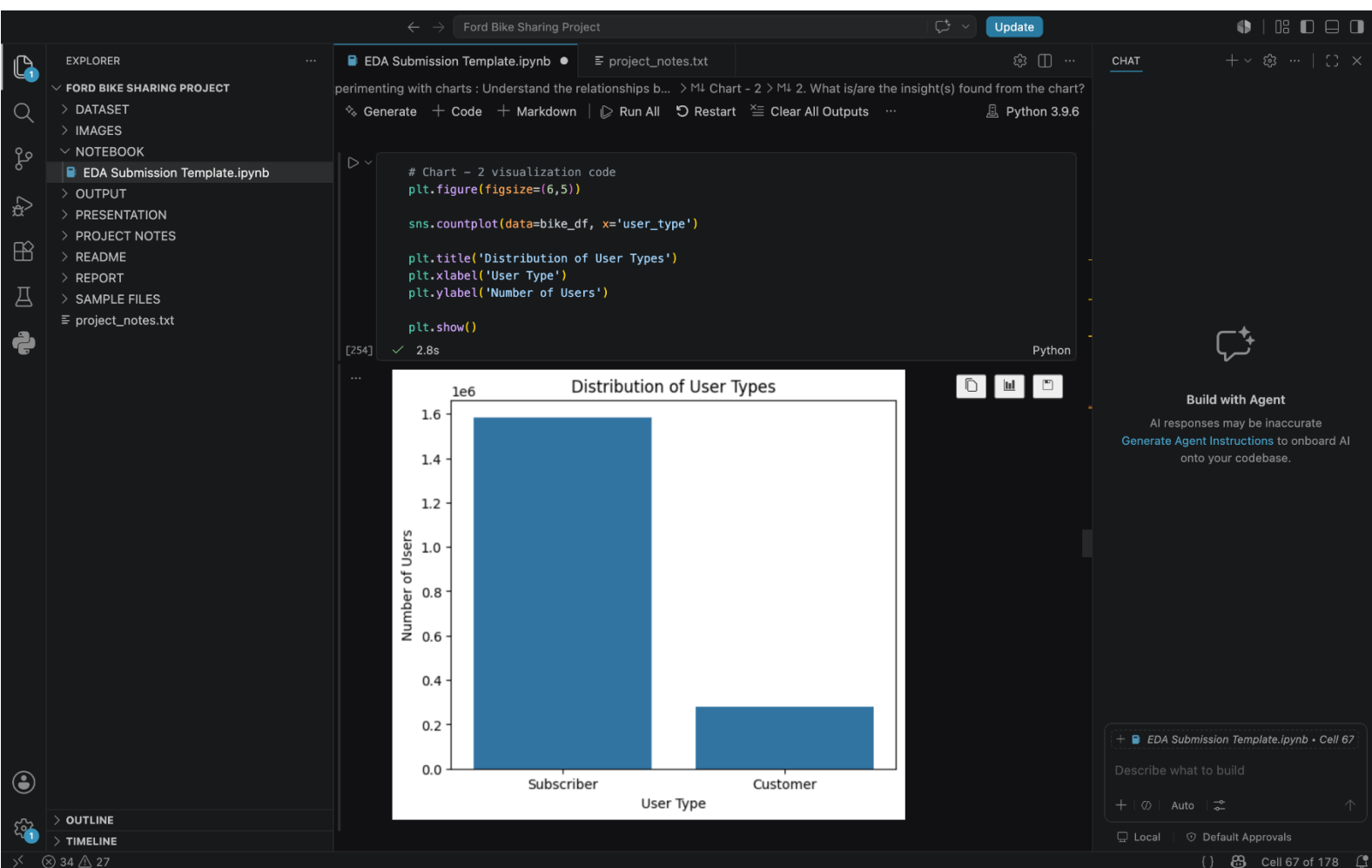


Figure 1: Data Cleaning Process

The dataset was examined for missing values, duplicate records, and incorrect data types. Date variables were converted into datetime format, and new variables such as Month, Day, Hour, and Age were created to support further analysis.

# USER & DEMOGRAPHIC ANALYSIS



**Figure 2: User Type Distribution**

## Insights

- Subscribers account for the majority of bike trips.
- Customers contribute a smaller share of total rides.
- Subscription membership is a key driver of service usage.

# GENDER DISTRINUTION

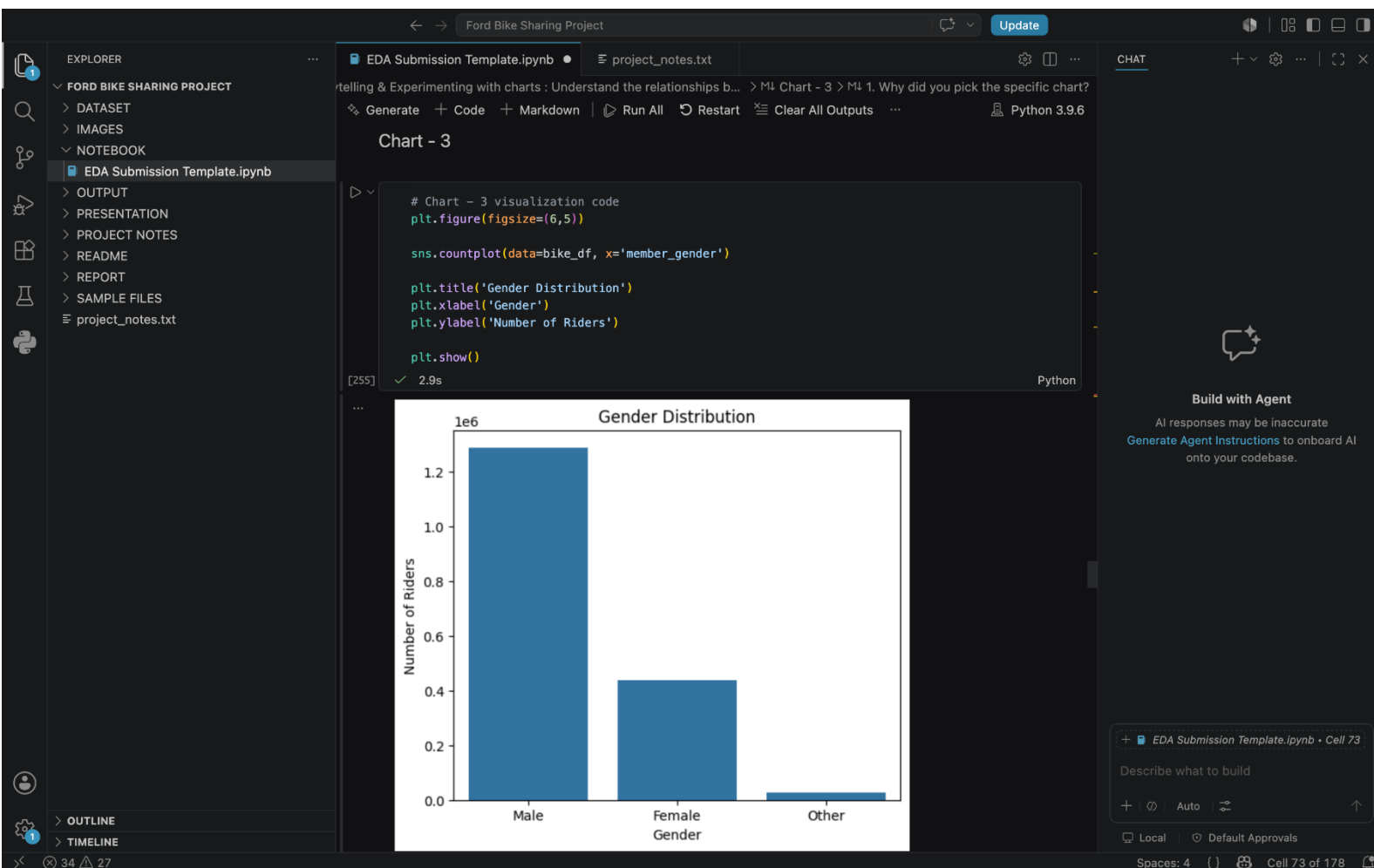
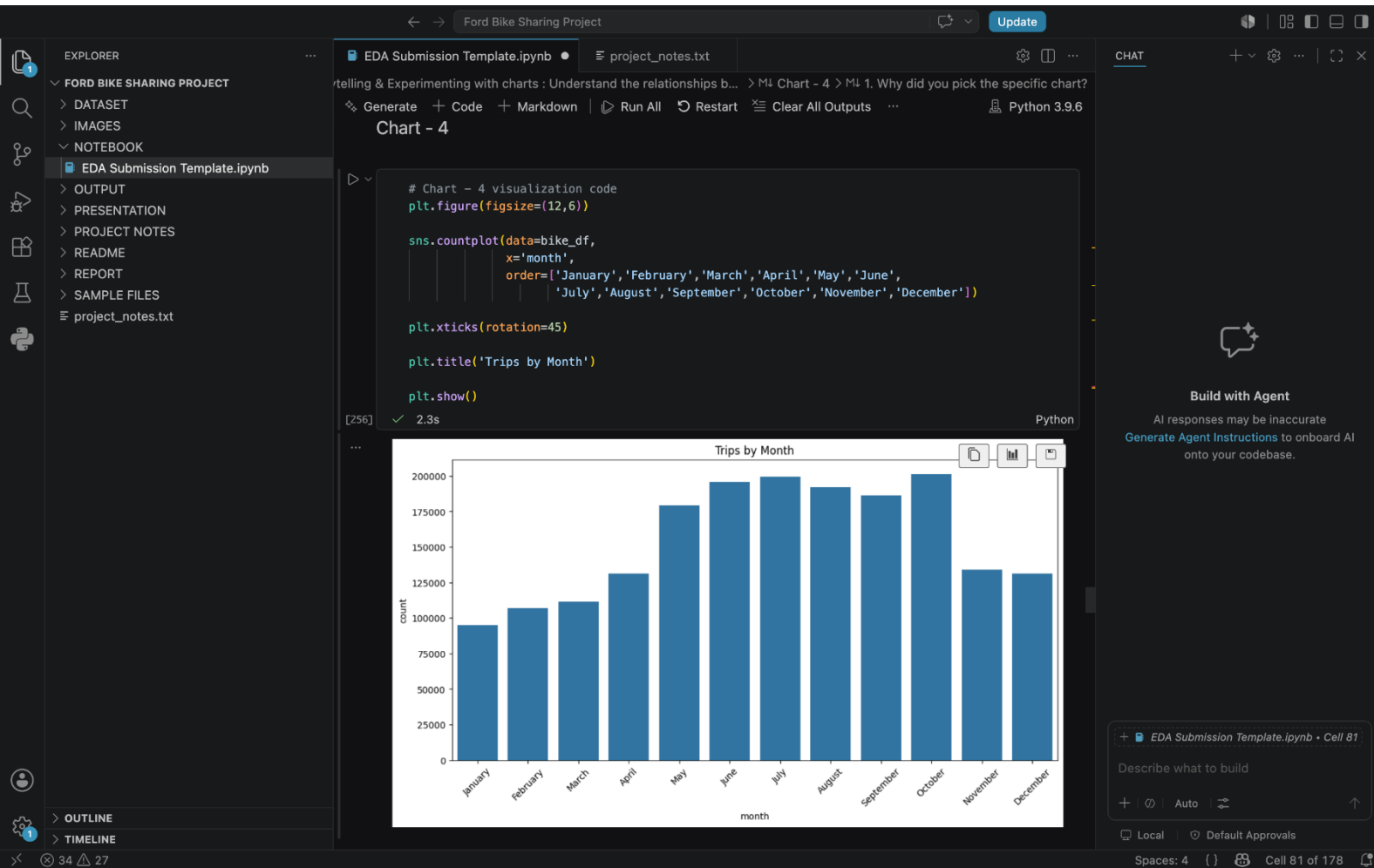


Figure 3: Gender Distribution

## Insights

- Male riders account for the highest number of trips.
- Female riders also contribute significantly to bike usage.
- Gender distribution helps understand customer demographics.

# TIME ANALYSIS



**Figure 4: Monthly Trip Distribution**

## Insights

- Bike usage varies across months.
- Seasonal trends influence customer demand.
- Demand forecasting can improve operational planning.

# TRIPS BY HOUR ANALYSIS

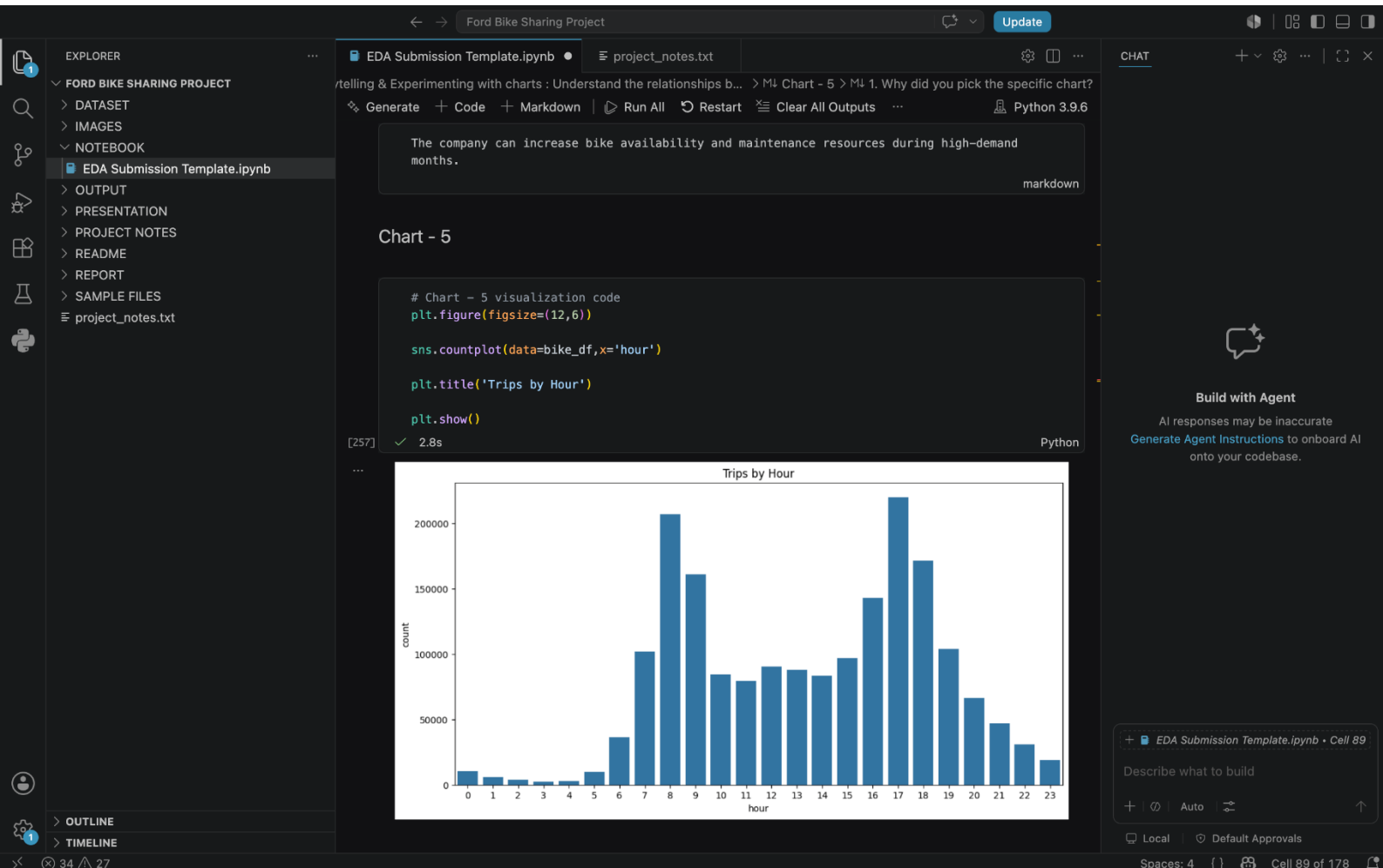


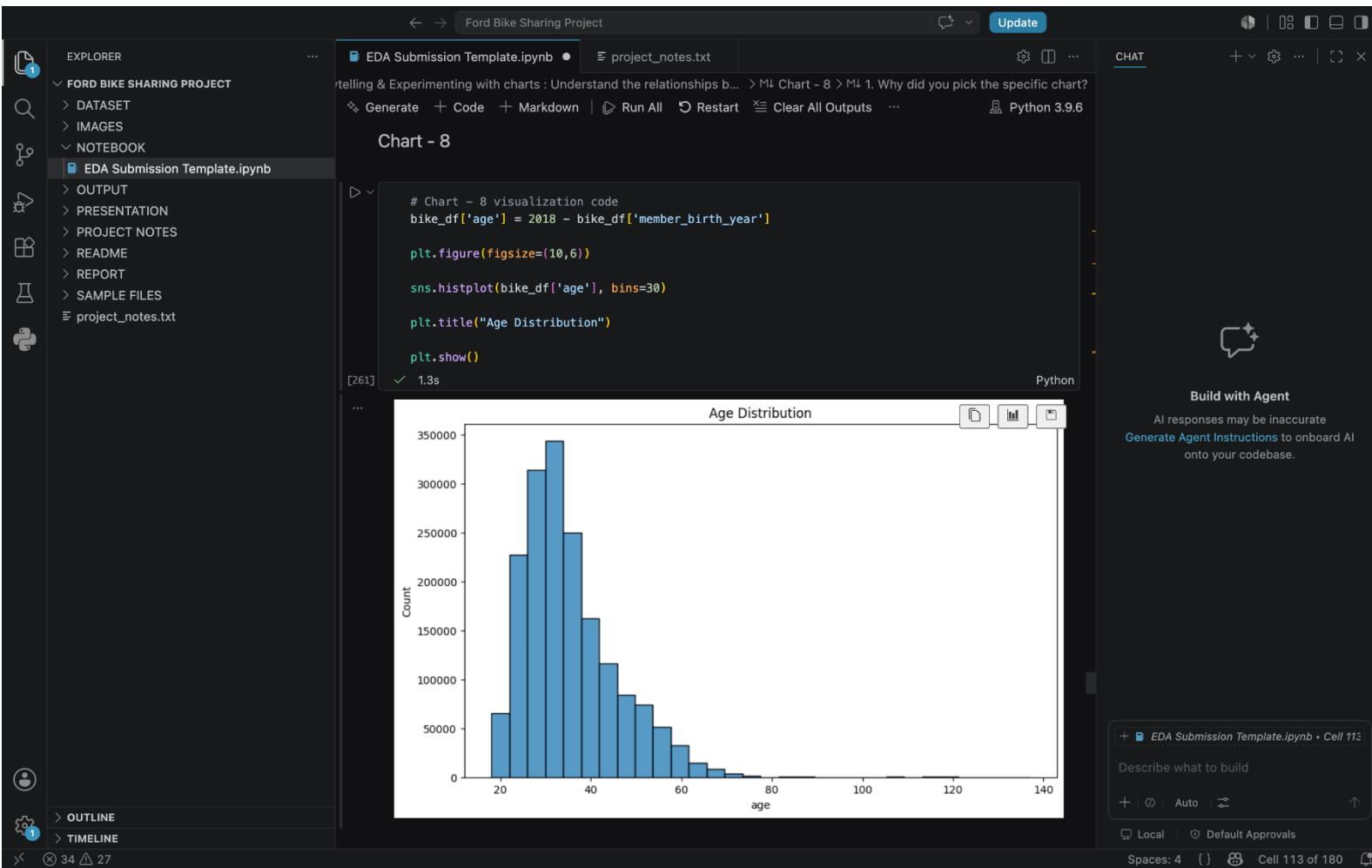
Figure 5: Trips by Hour

## Insights

- Peak demand occurs during commuting hours.
- Most rides are associated with daily travel needs.
- Late-night demand is comparatively lower.

# AGE ANALYSIS

## AGE DISTRIBUTION



**Figure 6: Rider Age Distribution**

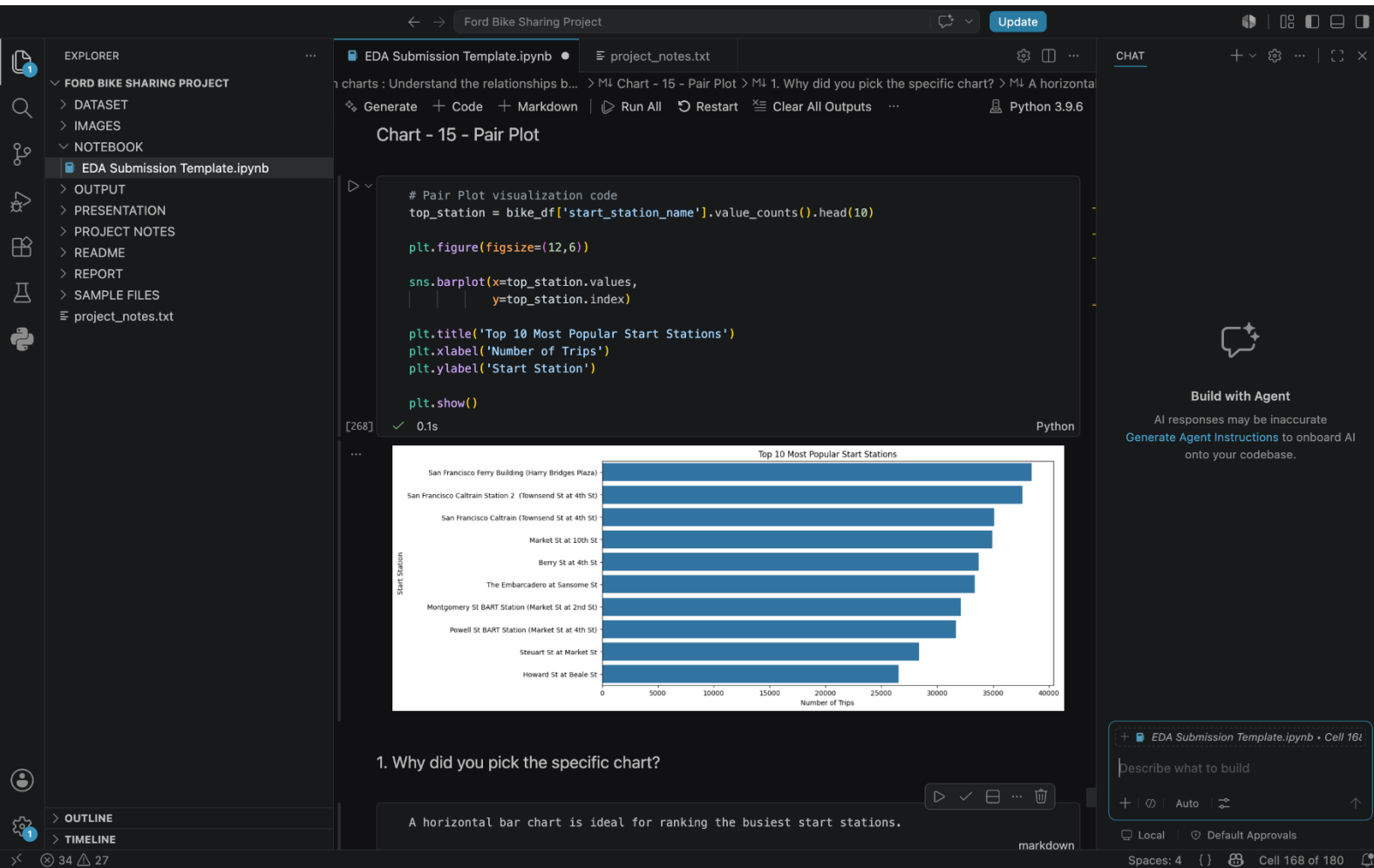
## INSIGHTS

- Most riders belong to young adult age groups.
- Working professionals form a major portion of users.
- Age-based segmentation can support targeted marketing.



# STATION ANALYSIS

## POPULAR STATIONS ANALYSIS



## INSIGHTS

- A few stations generate the majority of trips.
- High-demand stations require efficient bike allocation.
- Station popularity helps optimize resource planning.

## KEY FINDINGS

- Subscribers generate the majority of total trips.
- Most rides are short in duration.
- Peak demand occurs during weekday commuting hours.
- Young adults form the largest rider segment.
- A small number of stations account for the highest usage.
- Bike usage varies across months and seasons.
- Customer behavior patterns support data-driven operational decisions.

## RECOMMENDATIONS

- Increase bike availability during peak commuting hours.
- Expand bike capacity at high-demand stations.
- Introduce offers to convert customers into subscribers.
- Promote the Bike Share for All program.
- Optimize maintenance schedules during low-demand periods.
- Use seasonal demand trends for resource planning.
- Continuously monitor customer behavior using analytics.

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## CONCLUSION

The Ford GoBike System Data Exploration project successfully transformed raw trip data into meaningful business insights through exploratory data analysis. The analysis identified key usage patterns, customer demographics, peak demand periods, and station popularity trends.

The findings reveal that subscribers contribute the majority of trips, most rides are short and commute-oriented, and demand is concentrated around specific stations and time periods. These insights can support better operational planning, resource allocation, customer engagement, and strategic decision-making.

Overall, the project demonstrates the effectiveness of data analytics in understanding customer behavior and improving transportation services through evidence-based recommendations.

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